

Mobile Phone Price Range Classification Using Machine Learning

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1. Abstract

This project analyzed the factors that influence smartphone prices. We did this by classifying and separating devices into price categories based on their technical specifications. Using the “Mobile Price Classification” dataset, we addressed the challenge of determining appropriate price ranges in a highly competitive global market. Features like RAM, battery capacity, and processor speed interact in complex ways with said market. This study treated the problem as a classification task. We implemented and compared four machine learning models, those being: K Nearest Neighbours, Neural Network, Decision Tree and Random Forest. To make sure we had methodological correctness and fair comparison for all the models, they were all trained and evaluated using a standardized train-test split. They were assessed based on accuracy, precision, recall, F1-score, and ROC-AUC. We identified the most effective model for price prediction and gained insights into which hardware attributes/specifications serve as the primary drivers of smartphone cost.

2. Introduction

The smartphone industry worldwide has grown very competitive with the fast rise of the Chinese players, including Xiaomi, Oppo, Vivo, and Realme. These firms have broken the conventional pricing model of the smartphone market through the sale of products with premium specifications, including large RAM,

high-end processors, sophisticated cameras, and high-resolution screens at very low prices compared to the popular brands, such as Apple and Samsung. This change has escalated competition, and it is now more difficult to fix the right price bracket in new devices for the manufacturer and analysts. The current prices of smartphones are determined by a sophisticated blend of technical aspects such as RAM, built-in memory, battery life, processor speed, camera quality, and resolution of the display. The nature of interactions between these attributes and the dynamism of the market is such that it is no longer easy to estimate what price category a device should belong to. Statistically driven methods can be used to reveal trends between technical specifications and the price of the products. The machine learning models that can be used to categorize smartphones based on their prices can thus give some interesting insights into the impact of the features of the device on the price and how companies can better position their products in a highly dynamic market.

Past research has used machine learning techniques to categorize or estimate the price ranges of mobile phones based on the phone specifications. Common supervised learning algorithms used by researchers to tackle classification problems with structured tabular data include Logistic Regression, Decision Trees, Support Vector Machines, Random Forests and K-Nearest Neighbors. Such strategies were able to

elicit associations between hardware characteristics and product pricing typologies. Nevertheless, previous studies tended to concentrate on the accuracy of models, without a consistent evaluation of models through several measures or a clear description of how preprocessing choices were made. Moreover, not all studies had a standardized experimental design, and it is challenging to compare models fairly. Thus, a transparent and replicable workflow that compares a number of machine learning models within a similar preprocessing and evaluation pipeline is still useful to learn which models are better to use to classify smartphone prices.

In this project, we created and compared several supervised machine learning models to categorize smartphones in price ranges, training and testing them under identical conditions. This project aims to find out which machine learning model will be the most effective in predicting the price category of phones and to examine which features of a phone are the most predictive of the price category.

3. Method

3.1 Dataset Description

The *Mobile Price Classification* dataset from Kaggle was used for model training and evaluation. The dataset consisted of 2,000 mobile phone instances, each described by numerical and categorical attributes and a corresponding price-range class label. These attributes included hardware specifications, connectivity indicators, camera features, and physical characteristics of the devices.

3.2 Data Splitting

We used a fixed train-test split (75% train, 25% test) for our data, which all models used for fair comparison between their performances. A random seed was used whilst performing this split for consistency. To avoid data leakage, we made sure any preprocessing steps were only performed on the training data and that the same parameters

applied to the training set were applied in the same way for the testing set.

3.3 Preprocessing

We implemented preprocessing by:

- **Categorical encoding:** for some of the models, we used one-hot encoding for categorical variables
- **Quality checks:** We detected any outliers present in the data and confirmed that the labels were well-distributed.
- Standardising the data so that models like a K Nearest Neighbour model were able to perform optimally.
- There were no missing values present in the dataset, so there was no need to handle those cases.

3.4 Models

We used and compared the following models to reach our goal:

- **K Nearest Neighbour:** A model which classifies a data point based on its nearest neighbours
- **Decision Tree:** A model which classifies data points based on multiple decision nodes
- **Neural Network:** A model which will be able to predict each classification through layers of nodes
- **Random Forest:** A model that uses multiple decision trees, taking the most popular classification of a data point

3.5 Evaluation metrics

To evaluate the model performance comprehensively, so we reported:

- **Accuracy:** How many of each model's predictions were the correct prediction, positive or negative
- **Precision:** How many of its predicted positive labels were actually positive
- **Recall:** How many of the actual positive labels were predicted by each model

- **F1-score:** A balance between precision and recall
 - **ROC-AUC:** ranking a model's quality based on different thresholds
- We used these metrics per price classification. The same was done when making confusion matrices.

3.6 Success criteria

We defined success as improved classification performance compared to the K Nearest Neighbour model, which was used as a benchmark. We also emphasised:

- Decision nodes from the Decision Tree, to see which features were vital towards a phone's price
- Important features identified by the random forest for the same effect
- The f1-score for each classification, to strike a balance between precision and recall

4. Results

Table 1 summarises the performance of each model

Table 1
Overall performance of each model

Model	Accuracy	Precision	Recall	F1
K Nearest Neighbour	0.64	0.66	0.64	0.64
Decision Tree	0.81	0.82	0.81	0.81
Random Forest	0.91	0.92	0.91	0.91
Neural Network	0.93	0.93	0.93	0.93

The Neural Network performed best across all models, with the Random Forest very close in performance. The Decision Tree performed well as well, and the K-Nearest Neighbour model performed the worst of the four models. Figure 1 shows the confusion matrix of the best-performing model, the Neural Network.

Figure 1
Confusion Matrix of the Neural Network

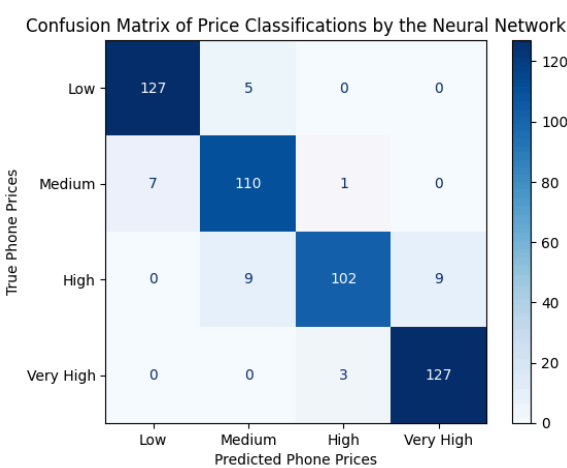


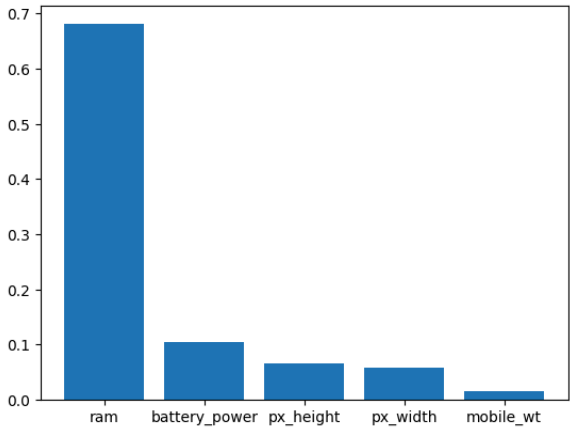
Table 2 shows a class-by-class performance of the Neural Network.

Table 2
Performance of Neural Network by class label

Price Range	Accuracy	Precision	Recall	F1
Low	0.98	0.95	0.96	0.95
Medium	0.96	0.89	0.93	0.91
High	0.96	0.96	0.85	0.90
Very High	0.98	0.93	0.93	0.95

From Figure 2, we can see that the Random Forest identified that a phone's RAM size is the most important feature when predicting the price of a phone.

Figure 2
Feature importances from the Random Forest



5. Discussion

The four models showed a clear performance ranking. The Neural Network achieved the highest overall performance, followed closely by the Random Forest, while the Decision Tree performed moderately well and K-Nearest Neighbour performed worst. This suggests that more flexible nonlinear models were better able to capture the relationship between phone specifications and price range.

K-Nearest Neighbour likely performed weakest because it relies directly on distance in feature space. Although feature scaling was applied, the middle price classes likely overlapped in their specifications, making distance-based classification less effective. The Decision Tree improved on KNN by learning clearer feature-based splits, but as a single tree it was still less robust than the stronger models.

The Random Forest performed very strongly, likely because combining multiple decision trees allowed it to capture nonlinear feature interactions while reducing the instability of a single tree. The feature-importance results also showed that RAM was the most influential predictor of smartphone price range, which is reasonable because memory capacity is closely related to device performance and market positioning. The Neural Network achieved the best results overall, suggesting that the relationship between phone specifications and price range was complex and better captured by a layered nonlinear model.

The *Mobile Price Classification* dataset from Kaggle was used for model training and evaluation. The dataset consisted of 2,000 mobile phone instances, each described by numerical and categorical attributes and a corresponding price-range class label. These attributes included hardware specifications, connectivity indicators, camera

features, and physical characteristics of the devices.

The study had some limitations. It used only one public dataset, hyperparameter tuning was limited, and the best-performing Neural Network was less interpretable than the tree-based models. Future work could include stronger tuning, testing on additional datasets, and using more interpretability methods for the Neural Network.

6. Data Availability

The dataset used in this study is publicly available and cited in the references. The project code is available [here](#).

7. Author Contributions

Armaan Gill - Neural Networks

Balraj Saran - Decision Trees

Joseph Odihi - Preprocessing, Random Forest and final report

Hassan Adnan - K Nearest Model, evaluation, insights and final report

8. References

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